

TTT4295 - Acoustic Signal Processing

Exercise 1 - Signal processing basics

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All the tasks are mandatory and should be handed in for approval (online).
The deadline is specified in Canvas.

Basics

1 The complex signal representation form

A microphone has a sensitivity of 50 mV/Pa.

a) If a sound pressure signal with the following expression is measured:

$$p(t) = 0,8e^{j(\omega t + \pi/4)} \text{ Pa}$$

What is the electric output signal from the microphone, expressed on the real-world form?

b) If the microphone produces a real electric output signal of:

$$e_{\text{out}}(t) = 30 \sin(\omega t + 1,2) \text{ mV}$$

How can the corresponding sound pressure be written in complex form?

2 dB-values vs. rms-values

a) We have measured a noise signal and filtered it into octave bands. This has given us the following RMS-values for the sound pressure:

Octave band:	125	250	500	1k	2k	4k	[Hz]
p_{rms} :	80	50	30	25	20	20	[mPa]

a) **What is the total (unfiltered/broadband) RMS-value of the signal?**

Hint: Remember that one should add the rms-values ****squared****.

b) **What is the sound pressure level in each octave band?**

Hint: Check the definition of sound pressure level.

c) Calculate the sound pressure level for the unfiltered/broadband sound signal in two ways:

1. **By using the definition of the sound pressure level for the total rms-value from task (a).**
2. **By using the uncorrelated addition of the dB-values in task (b) above.**

Result check: you should almost get the same result by the two methods in tasks (c) Rounding errors will cause a small difference with the two methods.

3 Averaging dB-values

We have measured the sound pressure level in six positions in a room and we want to find the representative average value.

Note: the representative average value is found by averaging p_{rms}^2 -values.

The six measured sound pressure levels were:

Meas. pos.:	1	2	3	4	5	6	
L_p :	71,6	72,3	72,5	70,9	70,1	72,0	[dB] re. 20 μPa

What is the representative average value?

Result check: you should get a value which is quite similar to, but not exactly the same as, the average of the dB-values.

4 A-weighting sound signals

We have (again) measured a noise signal and filtered it into octave bands. We now have the RMS-values of the electric signals. The microphone we used has a sensitivity of 50 mV/Pa, *and we used a microphone amplifier with 20 dB gain*. Then we measured these RMS-voltages (after the microphone amplifier had amplified the signals by 20 dB):

Octave band:	125	250	500	1k	2k	4k	[Hz]
e_{rms} :	70	40	40	30	8	8	[mV]

- a) **What are the sound pressure levels in the octave bands?**
- b) **What is the total (broadband) sound pressure level?**
- c) **What are the A-weighted sound pressure levels in the octave bands?**

Hint: You can find the A-weighting factor in dB for each octave band in the compendium.

- d) **What is the total A-weighted sound pressure level?**